

Dhananjay Nandedkar, DGM-marketing, Titan Energy Systems, feels that the National Solar Mission policy will unleash immense potential of the cells and modules market. "For example, 1 MW will need 4444 modules of 225 Wp size, so total of 20,000 MW will need 88,880,000 modules, which is a huge quantity," he adds.

Sunny states

Some sunny states of India are giving impetus to this scenario. Rajasthan, with 2200 hours of sunshine per year, has the plan to develop 50 MW photovoltaic (PV) power plants.

Gujarat recently launched a 500MW plan. Although the feed-in tariff (FIT) proposed by the government is not particularly attractive (Rs 15/kWh for 10 years), Indian PV project developers are grabbing land in the state. More than 3 GW of proposals have already been submitted. "Clearly, key players have recognised the long-term potential in the solar domain. Any city can become a solar city provided there is barren land with good sunshine for at least five hrs a day," states Nandedkar. Astonfield Renewable Resources will develop a 200MW PV farm in Gujarat. IT Power, Malaviya Energy Consultancy, Orb Energy, Solar India Online, Sripower and Signet Solar have all set their foot in the state.

Recently, at Solarcon India, Andhra Pradesh Chief Minister K. Rosaiah announced a solar city at Kadiri in Anantapur district, an exclusive cluster for setting up solar farms. Four firms—Sunborne Inc., Lanco Solar, AES Solar and Titan Energy—have proposed to set up solar power generation units totaling 2000 MW with an investment of Rs 30,000 million in the solar city.

Role of the biggies

Realising the growing potential of solar power generation, there has been a flurry of announcements by companies to join the solar industry, with many already setting up their manufacturing plants in India.

"After so many years, India still

lags behind in basic manufacturing but is likely to pick up in the near future despite the high cost factor involved in manufacturing cells and modules," states Ajay Prakash Srivastava, president, Solar Energy Society of India.

The upsurge in demand in India by about 25 per cent is prompting several players to pump in huge investments. Biggies like Solar Power India, Tata BP, Reliance Industries, Signet Solar, Applied Materials, Solar Semiconductor, Moser Baer, Orion Solar, Numeric and DD Berg have major plans for this industry. "This will give a big boost to the solar domain as these companies may invest a lot of money in R&D to compete in the global market, paving way for new and cheaper technology. This, in turn, will make solar energy accessible to common man. As more and more companies take to manufacturing in the solar domain, the cost of manufacturing is expected to reduce," says M.R. Rajesh, director, Powerone Micro System.

Moser Baer has announced that it is building a \$250-million plant to make solar energy products in Greater Noida, near New Delhi, using thin-film technology from Applied Materials. "Sales of about \$100 million annually are expected once the plant becomes fully operational," says Rajiv Arya, CEO-PV, Moser Baer.

Signet plans to conduct R&D in India and to reduce costs through product innovation. Tata BP Solar operates one of the oldest solar energy equipment manufacturing plants in India near Bengaluru. The plant is being expanded at an investment of \$100 million. US-based Cypress Semiconductor is reportedly considering a \$50 million facility to make solar cells and wafers near Hyderabad. SunTechnics Energy also has a manufacturing plant in Bengaluru and plans to expand it. The Michigan, US-based company, which has a manufacturing unit in Pune, intends to invest more in India.

Driving factors

One of the major factors that can drive the solar industry up the growth lad-

der is the telecom sector. According to Prasanna Gargava, vice president, business development, XL Telecom & Energy, solar power solutions for telecom towers can create new opportunities in India. Although the majority of telecom towers are presently backed-up by UPS and gensets, slowly and steadily many of them are opting for solar power solutions. This will be an environment-friendly, convenient and cost-effective way of overcoming traditional electricity crisis.

XL Telecom & Energy, working closely with Bharat Sanchar Nigam Limited (BSNL), has bagged a Rs 235-million order. It supplied 13,000 units of low-capacity-based standalone solar PV power to telecom applications and towers. "Moreover, we will supply more than 50,000 units of power to the new circles of BSNL. We are also in talks with other private telecom service providers, including Tata and Reliance," adds Gargava.

Tata BP, Moser Baer, DD Berg Project Consultants and Icomm Tele are some of the other players who are providing solar power to telecom towers. "Currently, we are generating solar power for Shyam Telecom in Gurgaon. Setting up small solar plants near telecom towers is not only the best option for smooth running of these towers but is lucrative business for solar players as well," says Shanker Grover, senior engineer, DD Berg Project Consultants.

"Every month, millions of new mobile phones are purchased in India. This is an interesting market segment for solar PV as thousands of new GSM poles will be needed across the country to provide connectivity," points out Nandedkar. "One-time investment, no recurring costs, less dependency on grid supply—which may not be available in remote locations, about 80 per cent accelerated depreciation benefit with huge savings on tax liability, savings on investment cost on UPS or genset and low maintenance are some of the benefits that can be reaped by opting solar power for telecom towers," he adds.

Manufacturing of PV cells and mod-